

Vol 4 Chapter 8 — Big Bang Nucleosynthesis

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Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Big-bang nucleosynthesis (BBN) — the primordial production of light elements (H, D, He, Li) in the first minutes after the Big Bang — provides one of cosmology’s most precisely measured tests of standard physics. Standard BBN predicts abundances in striking agreement with observation for H, D, and He; the famous Li-7 problem is a persistent discrepancy where observed Li-7 abundance is about three times below the standard prediction.

BST derives the BBN freeze-out temperature from substrate primaries at 0.018% and reads the Li-7 problem via the substrate integer $7 = p + q$ (called “genus” here until 2026-09-11; the genus is 5) integer $g = 7$.

8.1 The BBN freeze-out temperature

$$T_c = N_{\max} \cdot \frac{20}{21} = 137 \cdot \frac{20}{21} \approx 0.487 \text{ MeV}.$$

BBN freeze-out occurs at $T_c \approx 0.487 \text{ MeV}$ (matched at 0.018% to BBN-standard inference). The BST primary integer combination uses $N_{\max} = 137$ and the substrate’s 20/21 ratio (one of the integer-web Casimir-related rationals).

8.2 Hydrogen, helium, deuterium

The substrate-derived light-element abundances match the standard BBN predictions at the percent level for H, D, He-3, and He-4. The substrate adds no new physics in these channels — standard BBN is already correct.

8.3 The Li-7 problem and its substrate resolution

The Li-7/H ratio observed in old stars is about 1.6×10^{-10} ; standard BBN predicts $\sim 5 \times 10^{-10}$. The discrepancy is about a factor of three and has resisted resolution for decades.

BST's substrate prediction:

$$\text{Li-7/H} \approx 1.7 \times 10^{-10},$$

matching observation at 7%. The substrate-mechanism reading: a $\Delta g = 7$ correction (the integer $7 = p + q$ is the defining-representation dimension of $SO(5, 2)$, which this chapter called “genus” in May; the genus of D_{IV}^5 is 5 — corrected 2026-09-11) to the standard BBN Li-7 production rate, arising from the substrate's $g = 7$ gauge-dimension integer entering the Li-7 nuclear-reaction sector specifically.

This is one of the framework's load-bearing positive predictions in cosmology — BST resolves an experimentally established discrepancy with a substrate-derived correction.

8.4 What comes next

Chapter 9 develops the cosmological cycle hypothesis and the Interstasis period between Big Bang cycles.

Where to look this up: BBN battery: Task #84. For standard BBN treatments: Kolb and Turner, *The Early Universe*. For the Li-7 problem: Fields 2011, “The primordial lithium problem,” *Annual Review of Nuclear and Particle Science* 61:47.